APPENDIX C: CONVEX APPROXIMATIONS FOR UAV TRAJECTORY OPTIMIZATION

In this appendix, we detail the derivation of the convex approximations used in the trajectory subproblem (P3).

1 Concave Lower Bound of the Legitimate Rate

For UAV u communicating with BS b, define

$$X_{i,b}[n] \triangleq \|\mathbf{g}_i[n] - \mathbf{w}_b\|_2^2 . \tag{110}$$

Its legitimate spectral efficiency can be written as

$$C_{u,b}^{\text{leg}}[n] = F_{1,u,b}(\mathbf{X}_b[n]) - F_{2,u,b}(\mathbf{X}_b[n]), \tag{111}$$

where

$$F_{1,u,b}(\mathbf{X}_b[n]) = \log_2 \left(\sigma_b^2 + \sum_{i \in \mathcal{U}} \frac{p_i[n]\rho_0}{X_{i,b}[n]} \right), \tag{112}$$

$$F_{2,u,b}(\mathbf{X}_b[n]) = \log_2 \left(\sigma_b^2 + \sum_{\substack{i \in \mathcal{U} \\ i \neq u}} \frac{p_i[n]\rho_0}{X_{i,b}[n]} \right). \tag{113}$$

At the local trajectory point $\mathbf{Q}^{(\ell)}$, let

$$X_{i,b}^{(\ell)}[n] = \left\| \mathbf{g}_i^{(\ell)}[n] - \mathbf{w}_b \right\|_2^2 . \tag{114}$$

Since $F_{1,u,b}$ is convex with respect to $\mathbf{X}_b[n]$, its first-order expansion provides a global affine lower bound:

$$\begin{aligned} \underline{F}_{1,u,b}^{(\ell)}[n] =& F_{1,u,b}\Big(\mathbf{X}_b^{(\ell)}[n]\Big) \\ &+ \sum_{j \in \mathcal{U}} \xi_{u,b,j}^{(\ell)}[n] \left(X_{j,b}[n] - X_{j,b}^{(\ell)}[n] \right). \end{aligned} \tag{115}$$

where

$$\xi_{u,b,j}^{(\ell)}[n] = -\frac{p_j[n]\rho_0}{\ln 2 \left(X_{j,b}^{(\ell)}[n] \right)^2 \left(\sigma_b^2 + \sum_{i \in \mathcal{U}} \frac{p_i[n]\rho_0}{X_{i,b}^{(\ell)}[n]} \right)}. \tag{116}$$

To handle the interference term, introduce auxiliary squared-distance variables $\eta_{i,b}[n]$ and define $\boldsymbol{\eta}_b[n] = [\eta_{1,b}[n], \ldots, \eta_{U,b}[n]]^T$. The convex squared distance has the lower bound

$$\underline{X}_{i,b}^{(\ell)}[n] \triangleq X_{i,b}^{(\ell)}[n] + 2\big(\mathbf{g}_i^{(\ell)}[n] - \mathbf{w}_b\big)^T \big(\mathbf{g}_i[n] - \mathbf{g}_i^{(\ell)}[n]\big), \tag{117}$$

which satisfies $\underline{X}_{i,b}^{(\ell)}[n] \leq X_{i,b}[n]$. Due to $0 < \eta_{i,b}[n] \leq X_{i,b}[n]$, and therefore

$$F_{2,u,b}(\mathbf{X}_b[n]) \leq F_{2,u,b}(\boldsymbol{\eta}_b[n]). \tag{118}$$

On this basis, combining the two bounds yields

$$\underline{C}_{u,b}^{(\ell)}[n] \triangleq \underline{F}_{1,u,b}^{(\ell)}[n] - \log_2\left(\sigma_b^2 + \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} \frac{p_i[n]\rho_0}{\eta_{i,b}[n]}\right). \tag{119}$$

2 Convex Upper Bound of the Eavesdropping Rate

To handle the sensing-induced position uncertainty, introduce the slack variables $S_{u,e}[n] > 0$ and $Z_{i,e}[n] > 0$, satisfying

$$S_{u,e}[n] \le \left(\|\mathbf{g}_u[n] - \hat{\mathbf{w}}_e[n]\|_2 - r_e[n]\right)^2, \tag{120}$$

$$Z_{i,e}[n] \ge \left(\|\mathbf{g}_i[n] - \hat{\mathbf{w}}_e[n]\|_2 + r_e[n]\right)^2, \quad i \neq u. \tag{121}$$

They yield the conservative channel bounds

$$h_{u,e}^{\mathrm{up}}[n] \le \frac{\rho_0}{S_{u,e}[n]}, \qquad h_{i,e}^{\mathrm{low}}[n] \ge \frac{\rho_0}{Z_{i,e}[n]}. \tag{122}$$

Constraint (121) is convex. To approximate Equ.(120), define

$$d_{u,e}^{(\ell)}[n] = \left\|\mathbf{g}_u^{(\ell)}[n] - \hat{\mathbf{w}}_e[n]\right\|_2. \tag{123}$$

For $d_{u,e}^{(\ell)}[n] > r_e[n]$, a first-order lower bound is

$$\begin{aligned}\underline{G}_{u,e}^{(\ell)}[n] = &\left(d_{u,e}^{(\ell)}[n] - r_e[n]\right)^2 \\ &+ \frac{2\left(d_{u,e}^{(\ell)}[n] - r_e[n]\right)}{d_{u,e}^{(\ell)}[n]}\left(\mathbf{g}_u^{(\ell)}[n] - \hat{\mathbf{w}}_e[n]\right)^T \\ &\times \left(\mathbf{g}_u[n] - \mathbf{g}_u^{(\ell)}[n]\right).\end{aligned} \tag{124}$$

Thus, Equ.(120) is conservatively replaced by

$$S_{u,e}[n] \le \underline{G}_{u,e}^{(\ell)}[n]. \tag{125}$$

Using the slack variables, define

$$\Upsilon_{u,e}(\mathbf{Z}_e[n]) \triangleq I_e^{\mathrm{AN}}[n] + \sigma_e^2 + \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} \frac{p_i[n]\rho_0}{Z_{i,e}[n]}. \tag{126}$$

where $I_e^{\mathrm{AN}}[n] \triangleq \sum_{b\in\mathcal{B}} \Phi_{b,e}^{\mathrm{low}}[n]$.

Then,

$$G_{1,u,e} = \log_2\left(\Upsilon_{u,e}(\mathbf{Z}_e[n]) + \frac{p_u[n]\rho_0}{S_{u,e}[n]}\right), \tag{127}$$

$$G_{2,u,e} = \log_2(\Upsilon_{u,e}(\mathbf{Z}_e[n])). \tag{128}$$

The conservative eavesdropping rate is

$$R_{u,e}^{\mathrm{up}}[n] = a_u[n]\left(G_{1,u,e} - G_{2,u,e}\right). \tag{129}$$

At the local point $\mathbf{Z}_e^{(\ell)}[n]$, a global affine lower bound of $G_{2,u,e}$ is

$$\begin{aligned}\underline{G}_{2,u,e}^{(\ell)}[n] =& G_{2,u,e}\left(\mathbf{Z}_e^{(\ell)}[n]\right) \\ &- \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} \beta_{i,e}^{(\ell)}[n]\left(Z_{i,e}[n] - Z_{i,e}^{(\ell)}[n]\right).\end{aligned} \tag{130}$$

where

$$\beta_{i,e}^{(\ell)}[n] = \frac{p_i[n]\rho_0}{\ln 2\,\left(Z_{i,e}^{(\ell)}[n]\right)^2 \Psi_{u,e}^{(\ell)}[n]}, \tag{131}$$

and

$$\Psi_{u,e}^{(\ell)}[n] = I_e^{\mathrm{AN}}[n] + \sigma_e^2 + \sum_{\substack{i\in\mathcal{U}\\ i\neq u}} \frac{p_i[n]\rho_0}{Z_{i,e}^{(\ell)}[n]}. \tag{132}$$

Therefore, a convex upper approximation of the eavesdropping rate is

$$\overline{R}_{u,e}^{(\ell)}[n] = a_u[n]\left[G_{1,u,e} - \underline{G}_{2,u,e}^{(\ell)}[n]\right]. \tag{133}$$